

Frequency Analysis and Cryptanalysis of Vigenère Cipher

Plaintext Used: Hello, this is a sample sentence that I am using for frequency analysis. I will use this sentence to analyze the frequency of characters and identify useful patterns. At the same time, I will attempt to recover the original plaintext from the ciphertext to evaluate the vulnerability of the cipher to cryptanalysis attacks.

1. Frequency Analysis

The plaintext was encrypted using the Vigenère cipher and the resulting ciphertext was analyzed using the Frequency Test component in CrypTool 2. The frequency graph displays the percentage occurrence of individual characters in the ciphertext.

Unlike the Caesar cipher, Vigenère uses multiple alphabetic shifts based on a repeating key. Therefore, the same plaintext letter can be encrypted as different ciphertext letters. This spreads the frequency of common plaintext letters across several ciphertext characters, making a simple one-to-one frequency comparison less effective.

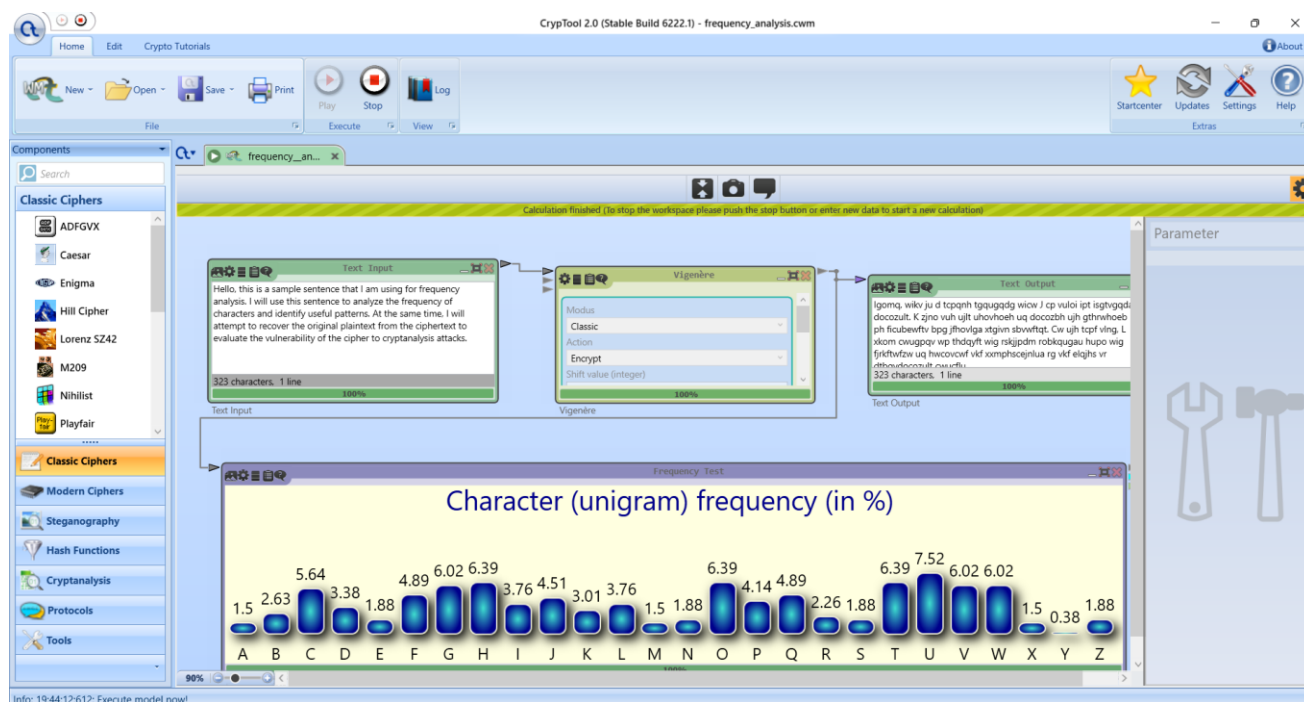


Figure 1: Vigenère ciphertext and character-frequency analysis in CrypTool 2.

2. Cryptanalysis Attack

The frequency distribution in Figure 1 does not directly reveal a single Caesar-like shift. This is because the Vigenère cipher uses different shifts for different positions of the plaintext. Consequently, an attacker cannot simply select the most frequent ciphertext letter and map it to E as was done in the Caesar attack.

A Vigenère cryptanalysis instead requires investigation of the repeating-key structure, such as possible key length and the frequency patterns associated with individual key positions. After the appropriate decryption settings are identified, the ciphertext can be decrypted and the original plaintext can be recovered.

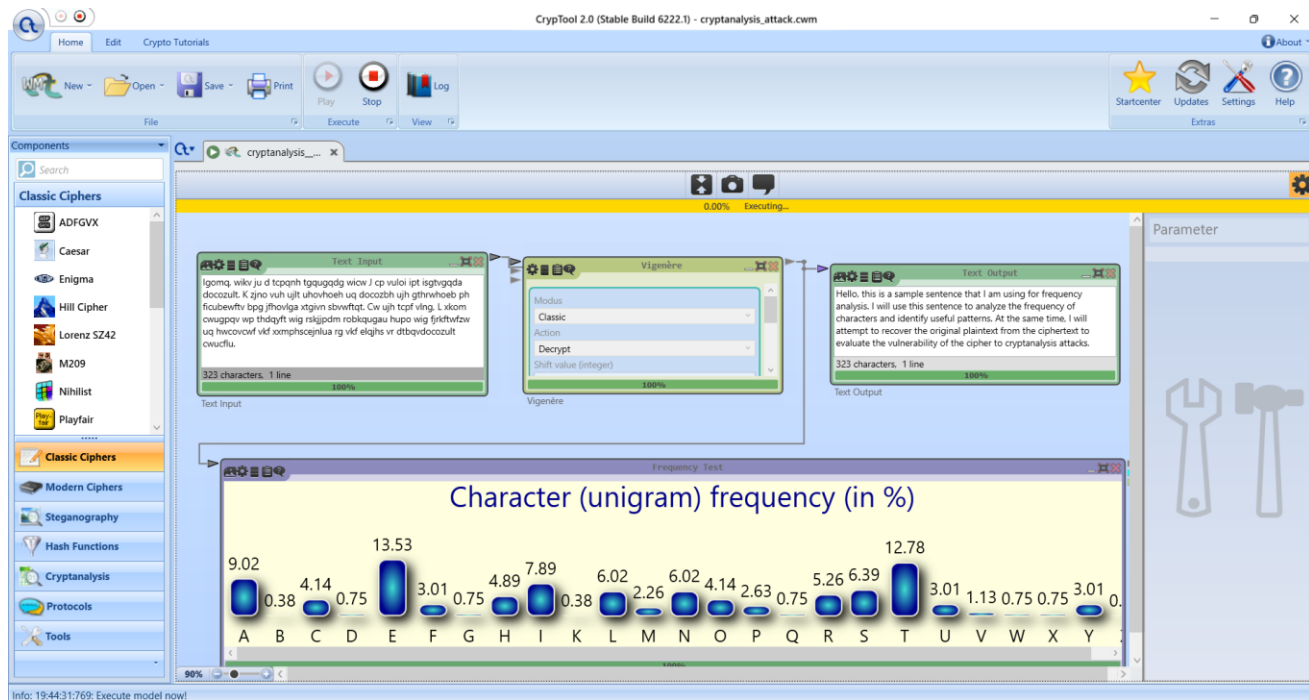


Figure 2: Vigenère decryption in CrypTool 2 recovering the original plaintext.

3. Observation

- The Vigenère cipher produces a more distributed frequency pattern than the Caesar cipher.
- A simple frequency analysis does not directly reveal one fixed encryption shift.
- The repeating-key structure provides information that can be used for cryptanalysis.
- With the correct decryption settings, CrypTool 2 recovers the original readable plaintext.
- Therefore, Vigenère is more resistant to simple frequency analysis than Caesar, but it is still vulnerable to classical cryptanalysis techniques.

4. Conclusion

The experiment demonstrates that Vigenère provides greater resistance to simple frequency analysis than the Caesar cipher because it uses multiple substitutions. However, its repeating-key structure can still be analyzed, especially when sufficient ciphertext is available. CrypTool 2 can be used to study these patterns and verify the recovery of the original plaintext.